UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence — Fall 2025
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Lecture 9

Content.

1. Triangle threshold in random graphs
2. Gaussian tail bounds
3. Hoeffding's lemma

1 Triangle threshold in random graphs

Theorem 1. *Consider $G(n, p_n)$, namely the Erdős–Rényi random graph on n vertices, where each edge appears with probability p, independently. Then*

$$\lim_{n\to\infty} \mathbb{P}\big[G(n,p) \text{ contains a triangle}\big] = \begin{cases} 0, & \text{if } p_n \ll \frac{1}{n}, \\ 1, & \text{if } p_n \gg \frac{1}{n}. \end{cases}$$

Proof of Theorem 1. Define

$$Z = \sum_{T \in \binom{[n]}{3}} I_T, \qquad I_T = \begin{cases} 1, & \text{if } T \text{ is a triangle}, \\ 0, & \text{otherwise}. \end{cases}$$

Here

$$\binom{[n]}{3} = \{\{i,j,k\} : 1 \le i < j < k \le n\}.$$

The expectation is

$$\mathbb{E}[Z] = \binom{n}{3}\mathbb{P}[I_T = 1] = \binom{n}{3} p_n^3 = \Theta(n^3 p_n^3).$$

Thus, if $p_n \ll 1/n$, then $\mathbb{E}[Z] = o(1)$ which by Markov's inequality implies that $G(n,p)$ doesn't contain a triangle whp as $n \to \infty$.

Suppose now $p_n \geq \frac{c_n}{n}$ with $c_n \to \infty$. Then $\mathbb{E}[Z] \to \infty$. We will combine this with a second moment bound.

Goal:

Show that

$$\mathrm{Var}(Z) = o\big((\mathbb{E}[Z])^2\big)\,.$$

This would imply

$$\mathbb{E}[Z^2] = (\mathbb{E}[Z])^2 + \mathrm{Var}(Z) = (\mathbb{E}[Z])^2\big(1 + o_n(1)\big).$$

$$\mathbb{P}[Z > 0] \geq \frac{(\mathbb{E}[Z])^2}{\mathbb{E}[Z^2]} = 1 - o_n(1) \quad \to 1 \quad \text{as } n \to \infty,$$

by Paley-Zygmund inequality, yielding the conclusion.
Note that

$$\mathrm{Var}(Z) = \sum_{T,T'} \mathrm{Cov}(I_T, I_{T'}),$$

where

$$\mathrm{Cov}(I_T, I_{T'}) = \mathbb{E}[I_T I_{T'}] - \mathbb{E}[I_T]\mathbb{E}[I_{T'}].$$

Here, (T, T') runs over $\binom{n}{3}^2$ pairs with $T, T' \in \binom{[n]}{3}$ and

$$|T \cap T'| = k.$$

The parameter $|T \cap T'|$ is referred to as the *overlap*.

Case 1: $k = 3$

$$\#\text{ of such pairs} = \binom{n}{3}$$

$$\mathbb{E}[I_T I_T] = \mathbb{E}[I_T^2] = \mathbb{E}[I_T]$$

$$\sum_{T,T'} \mathrm{Cov}(I_T, I_{T'}) \leq \binom{n}{3}\mathbb{E}[I_T] = \mathbb{E}[Z].$$

Case 2: $k = 2$

$$\#\text{ of such pairs} = \binom{n}{3} \cdot 3(n-3) = \Theta(n^4).$$

$$\mathbb{E}[I_T I_{T'}] - \mathbb{E}[I_T]\mathbb{E}[I_{T'}] = p^5 - p^6 = p^5(1-p).$$

So the whole contribution:

$$\sum_{\substack{T,T' \\ |T \cap T'|=2}} \mathrm{Cov}(I_T, I_{T'}) = O(n^4 p^5).$$

Case 3: $k = 1$

$$\#\text{ of pairs} = \binom{n}{3} \cdot 3\binom{n-3}{2} = \Theta(n^5).$$

$$\mathbb{E}[I_T I_{T'}] = p^6 \quad \Rightarrow \quad \mathrm{Cov}(I_T, I_{T'}) = 0.$$

Note: Even though $T \cap T' \neq \varnothing$, I_T and $I_{T'}$ are independent.

Case 4: $k = 0$

$$\mathrm{Cov}(I_T, I_{T'}) = 0.$$

$$\mathrm{Var}(Z) \leq \mathbb{E}[Z] + O(n^4 p^5) = \Theta(n^3 p_n^3) + O(n^4 p_n^5).$$

$$n^4 p_n^5 \leq n^4 p_n^4 = O(n^6 p_n^6) = o\big((\mathbb{E}[Z])^2\big), \quad \text{since } np_n \to \infty \text{ by assumption.}$$

□

Remark. *In some sense, our proof shows that the second moment is dominated by pairs (T, T') where $|T \cap T'| = \varnothing$. This is the essence of the second moment method; it is particularly prevalent in models such as this one where there is symmetry.*

2 Concentration inequalities

As a preliminary, consider coin tossing experiment. Let S_N be the number of heads in N tosses of an unbiased coin. Then:

$$\mathbb{E}[S_N] = \frac{N}{2}, \qquad \mathrm{Var}(S_N) = \frac{N}{4},$$

since

$$S_N = \sum_{i=1}^{N} B_i, \quad B_i \sim \mathrm{Bern}\big(\tfrac{1}{2}\big)\,.$$

Using this, we find

$$\begin{aligned}
\mathbb{P}\big[S_N \geq \tfrac{3N}{4}\big] &= \mathbb{P}\big[S_N - \tfrac{N}{2} \geq \tfrac{N}{4}\big] \\
&\leq \mathbb{P}\big[|S_N - \tfrac{N}{2}| \geq \tfrac{N}{4}\big] \\
&\leq \frac{\mathrm{Var}(S_N)}{(N/4)^2} \\
&= \frac{N/4}{N^2/16} \\
&= \frac{4}{N} \\
&= O\big(\tfrac{1}{N}\big)\,.
\end{aligned}$$

Question: Is the rate $O(1/N)$ optimal?

Heuristic:

$$\frac{S_N - N/2}{\sqrt{N/4}} \approx Z, \qquad Z \sim \mathcal{N}(0,1).$$

$$\mathbb{P}\big[S_N \geq \tfrac{3N}{4}\big] = \mathbb{P}\left[\frac{S_N - N/2}{\sqrt{N/4}} \geq \sqrt{N/4}\right] \approx \mathbb{P}[Z \geq \sqrt{N/4}].$$

Gaussian Tail Bound

Lemma 1. *For any $t > 0$,*

$$\frac{1}{\sqrt{2\pi}}\left(\frac{1}{t} - \frac{1}{t^3}\right) e^{-t^2/2} \;\leq\; \mathbb{P}[N(0,1) > t] \;\leq\; \frac{e^{-t^2/2}}{t\sqrt{2\pi}}.$$

Corollary 1. *If* $t > \frac{1}{\sqrt{2\pi}}$*, then*

$$\mathbb{P}[N(0,1) > t] \leq e^{-t^2/2}.$$

Corollary 2. *If* $t = \omega(1)$*, then*

$$\mathbb{P}[N(0,1) > t] = \frac{e^{-t^2/2}}{t\sqrt{2\pi}}\big(1 + o_t(1)\big).$$

Proof of Lemma 1. Recall that

$$\mathbb{P}[Z \geq t] = \int_t^\infty \frac{e^{-x^2/2}}{\sqrt{2\pi}}\, dx.$$

Using Mills' inequality,

$$\int_t^\infty \frac{e^{-x^2/2}}{\sqrt{2\pi}}\, dx \;\leq\; \int_t^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\left(\frac{x}{t}\right) dx = \frac{1}{t\sqrt{2\pi}} \int_t^\infty x e^{-x^2/2} dx.$$

Now note:

$$\int_t^\infty x e^{-x^2/2} dx = e^{-t^2/2},$$

so

$$\mathbb{P}[Z \geq t] \leq \frac{e^{-t^2/2}}{t\sqrt{2\pi}}.$$

Lower Bound

$$\int_t^\infty \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx = \int_t^\infty \frac{e^{-x^2/2}}{\sqrt{2\pi}} \cdot \frac{x}{x}\, dx.$$

Take

$$u = \frac{1}{x\sqrt{2\pi}}, \quad dv = x e^{-x^2/2} dx, \quad v = -e^{-x^2/2}.$$

$$\begin{aligned}
\int_t^\infty \frac{e^{-x^2/2}}{\sqrt{2\pi}} \cdot \frac{x}{x}\, dx &= \frac{1}{t\sqrt{2\pi}}\big[-e^{-x^2/2}\big]_t^\infty - \int_t^\infty \big(-e^{-x^2/2}\big)\left(-\frac{1}{x^2\sqrt{2\pi}}\right) dx \\
&= \frac{e^{-t^2/2}}{t\sqrt{2\pi}} - \int_t^\infty \frac{e^{-x^2/2}}{x^2\sqrt{2\pi}} dx \\
&= \frac{e^{-t^2/2}}{\sqrt{2\pi}}\left(\frac{1}{t} - \frac{1}{t^3}\right) + \int_t^\infty \frac{e^{-x^2/2}}{x^3\sqrt{2\pi}} dx \\
&\geq \frac{e^{-t^2/2}}{\sqrt{2\pi}}\left(\frac{1}{t} - \frac{1}{t^3}\right)
\end{aligned}$$

since $\int_t^\infty \frac{e^{-x^2/2}}{x^3\sqrt{2\pi}} dx \quad \geq 0.$ □

Remark: Iterating gives:

$$\frac{e^{-t^2/2}}{\sqrt{2\pi}}\left(\frac{1}{t}-\frac{1}{t^3}+\frac{c_5}{t^5}-\frac{c_7}{t^7}+\cdots\right).$$

Returning to coin tossing, this still doesn't suffice to recover an exponential rate. By Berry–Esseen theorem,

$$\left|\mathbb{P}[S_N \geq 3N/4] - \mathbb{P}[Z \geq \sqrt{N/4}]\right| = O\Big(\tfrac{1}{\sqrt{N}}\Big),$$

which does *not* recover the $e^{-\Theta(N)}$ rate.

Exercise: Let X_i be i.i.d. $\mathrm{Rad}\big(\frac{1}{2}\big)$. Define

$$S_N = \sum_{i=1}^{N} X_i.$$

Show that $\Theta(1/\sqrt{n})$ rate is necessary. *Hint:* Consider $\mathbb{P}[S_N = 0]$ and show that it is equal to $\binom{N}{N/2}2^{-N}$, which by Stirling's is $\Theta(1/\sqrt{N})$.

3 Hoeffding's Lemma

Recall that

$$\mathbb{E}[e^{sX}] = \int e^{sx}\, d\mathbb{P}_X.$$

Lemma 2 (Hoeffding's Lemma). *Let X be a random variable with $\mathbb{E}[X]$ such that $X \in [a,b]$ almost surely. Then*

$$\mathbb{E}[e^{sX}] \leq \exp\left(\frac{s^2(b-a)^2}{8}\right).$$

Proof I (worse constant, $8 \to 2$)

Lemma: Let X be a random variable such that $X \in [a,b]$ almost surely. Then

$$\mathrm{Var}(X) \leq \tfrac{1}{4}(b-a)^2.$$

This bound is attained when X takes a and b each with probability $1/2$.

Proof.

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Note that

$$\mathrm{Var}(X) = \inf_c \mathbb{E}[(X - c)^2],$$

which can be showing by considering the quadratic in c and taking derivatives. Taking $c = \frac{a+b}{2}$,

$$\mathrm{Var}(X) \leq \mathbb{E}\left[\left(X - \tfrac{a+b}{2}\right)^2\right].$$

Since $X \in [a, b]$,

$$\left|X - \tfrac{a+b}{2}\right| \leq \tfrac{b-a}{2}.$$

□

So

$$\mathrm{Var}(X) \leq \tfrac{1}{4}(b-a)^2.$$

Symmetrization Trick

Let $X' \overset{\text{iid}}{\sim} X$, with $\mathbb{E}[X'] = 0$.

$$X' - X \overset{d}{=} X - X'.$$

$$\mathbb{E}[e^{sX}] = \mathbb{E}\left[e^{sX - s\mathbb{E}[X']}\right].$$

By Jensen's inequality: If f is convex, then $\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$. If f is concave, then $\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$.
Here $f(x) = e^{-sx}$ is convex, so

$$\mathbb{E}[e^{sX}] \leq \mathbb{E}[e^{sX}] \cdot \mathbb{E}[e^{-sX'}].$$

Since X and X' are independent,

$$\mathbb{E}[e^{sX}] \leq \mathbb{E}[e^{s(X-X')}].$$

Now let $\varepsilon \sim \mathrm{Rad}\left(\frac{1}{2}\right)$ (a Rademacher random variable). Since $(X - X') \overset{d}{=} (X' - X)$,

$$\mathbb{E}_{X,X'}\left[e^{s(X-X')}\right] = \mathbb{E}_{X,X',\varepsilon}\left[e^{s\varepsilon(X-X')}\right].$$

$$= \mathbb{E}_{X,X'}\Big[\mathbb{E}_\varepsilon\left(e^{s\varepsilon(X-X')}\right)\Big].$$

Claim:

$$\mathbb{E}_\varepsilon[e^{s\varepsilon}] \le \exp\left(\tfrac{s^2}{2}\right).$$

If the claim holds, then

$$\mathbb{E}[e^{sX}] \;\le\; \mathbb{E}_{X,X'}\left[\exp\left(\tfrac{s^2}{2}(X-X')^2\right)\right].$$

Since $|X - X'| \le b - a$ almost surely,

$$\mathbb{E}[e^{sX}] \;\le\; \exp\left(\frac{s^2(b-a)^2}{2}\right).$$

Proof of Claim

$$\mathbb{E}[e^{s\varepsilon}] = \tfrac{1}{2}e^s + \tfrac{1}{2}e^{-s}.$$

Expand in series:

$$= \tfrac{1}{2}\sum_{k=0}^{\infty}\frac{s^k}{k!} + \tfrac{1}{2}\sum_{k=0}^{\infty}\frac{(-s)^k}{k!}.$$

Odd terms cancel, so

$$= \sum_{\ell=0}^{\infty}\frac{s^{2\ell}}{(2\ell)!}.$$

Now observe

$$\frac{1}{(2\ell)!} \le \frac{1}{2^\ell\,\ell!},$$

so

$$\sum_{\ell=0}^{\infty}\frac{s^{2\ell}}{(2\ell)!} \le \sum_{\ell=0}^{\infty}\frac{1}{\ell!}\left(\frac{s^2}{2}\right)^\ell = \exp\left(\frac{s^2}{2}\right).$$

Thus

$$\mathbb{E}[e^{s\varepsilon}] \le \exp\left(\tfrac{s^2}{2}\right).$$

Second Proof of Hoeffding's Lemma

Define

$$\Psi(s) := \log \mathbb{E}[e^{sX}], \qquad \Psi(0) = 0.$$

We have

$$\Psi'(s) = \frac{\mathbb{E}[Xe^{sX}]}{\mathbb{E}[e^{sX}]},$$

assuming the order of derivative and expectation can be exchanged.

$$\Psi'(0) = \mathbb{E}[X] = 0.$$

Route: Calculate $\Psi''(s)$ and upper bound it.
This uses the idea of *tilting* (change of measure).
Fundamental Theorem of Calculus
Define a tilted measure:

$$dQ = \frac{e^{sX}}{\mathbb{E}_P[e^{sX}]}\, dP_X.$$

This is a valid probability measure since

$$\int dQ = \int \frac{e^{sX}}{\mathbb{E}_P[e^{sX}]}\, dP_X = \frac{\mathbb{E}_P[e^{sX}]}{\mathbb{E}_P[e^{sX}]} = 1.$$